Endpoint L^1 Vertical Inequalities in Nilpotency Step Two Verification, Strengthening, and Literature Status

June 18, 2026

Abstract

We verify the argument in the supplied solution packet for Ryoo's endpoint vertical-versus-horizontal question. The proof is correct after making two standard points explicit: the elementary Heisenberg subgroup is closed, and Weil's quotient integration formula is applied to that closed unimodular subgroup. We also strengthen its scope. The Carnot assumption is unnecessary: every connected, simply connected, step-two nilpotent Lie group equipped with an arbitrary bracket-generating left-invariant Carnot–Carathéodory structure satisfies the endpoint inequality with exponent $q = 4$ in every commutator direction. A quantitative constant is given by a natural factorization quasinorm of the central vector. Hence Ryoo's Question 38 is completely resolved in nilpotency step two, though the optimal exponent can be smaller than four. We did not find a published or preprint resolution, or a counterexample, for the general higher-step question, and the higher-step problem remains open on the evidence located in the literature search described below.

Packet status

This is a strengthened partial packet for Ryoo's endpoint L^1 vertical-versus-horizontal problem. It is a complete positive answer in nilpotency step two, including non-Carnot step-two groups equipped with an arbitrary bracket-generating left-invariant Carnot–Carathéodory structure. It is not a full solution of the higher-step endpoint problem.

1 The endpoint question

Let G be a connected, simply connected nilpotent Lie group, let $W \subseteq \mathfrak{g}$ be a bracket-generating subspace with a Euclidean norm, and let $\nabla_W f$ denote the corresponding left-invariant horizontal gradient. If v lies in the last nonzero term of the lower central series and the nilpotency step is s, Ryoo asks for which $q \geq 1$ one has

$$\left(\int_0^\infty \left(\frac{\int_G |f(g \exp(tv)) - f(g)| \, \mathrm{d}g}{t^{1/s}} \right)^q \frac{\mathrm{d}t}{t} \right)^{1/q} \lesssim_{G,q,v} \int_G |\nabla_W f(g)| \, \mathrm{d}g \tag{1}$$

for every $f \in C_c^\infty(G)$. The usual essential-supremum interpretation is used when $q = \infty$.

Question 38. *Let G be a nonabelian simply connected nilpotent Lie group of nilpotency step s, and let $v \in \underbrace{[G,[G,\cdots,G]]}_{s \text{ times}}$ be normalized so that $d_G(v,e_G)=1$. For which exponents $q \geq 1$ does every smooth and compactly supported $f : G \to \mathbb{R}$ satisfy the following?*

$$\left(\int_0^\infty \left(\int_G \left|f\left(h\exp(tv)\right)-f(h)\right| d\mu(h)\right)^q \frac{dt}{t^{1+q/s}}\right)^{1/q} \lesssim_{G,q} \int_G \|\nabla f(h)\|_{\ell_2^k} d\mu(h). \tag{26}$$

This tensorizes easily into an inequality for functions $f : G \to L^1$. Similarly, we may ask the following question.

Question 39. *Let Γ be a not virtually abelian finitely generated group of polynomial growth. Choose v_Γ and $s > 0$ as in Theorem 7. For which exponents $q \geq 1$ do there exist $c = c(\Gamma) \in \mathbb{N}$ such that for any $n \in \mathbb{N}$ and $f : \Gamma \to L^1$ we have the following?*

$$\left(\sum_{k=1}^{n^s} \frac{1}{k^{1+q/s}} \sum_{x \in B_n^\Gamma} \left\|f\left(xv_\Gamma^k\right)-f(x)\right\|_{L^1}^q\right)^{1/q} \lesssim_{\Gamma,q} \sum_{x \in B_{cn}^\Gamma} \sum_{a \in S} \|f(xa)-f(x)\|_{L^1}.$$

By the discretization argument of Section 3, an exponent q that answers Question 38 positively also answers Question 39 positively, when G is chosen as the Malcev completion of the torsion subgroup quotient of a finite index nilpotent subgroup of Γ.

Because the case $q = \infty$ formally holds for Question 38 (see Remark 29), by Hölder's inequality the set of q that satisfies Question 38 is either of the form $\left(q_{G,v},\infty\right)$ or $\left[q_{G,v},\infty\right)$ for some $q_{G,v} \in [1,\infty]$ depending on G. We make the following formal definition.

Definition 40. *Given a nonabelian nilpotent Lie group G, let $v \in \underbrace{[\mathfrak{g},[\mathfrak{g},\cdots,\mathfrak{g}]]}_{s \text{ times}}$ with $d_G\left(\exp(v),e_G\right)=1$. Let $q_{G,v}$ denote the infimum of those $q \geq 1$ such that* (26) *holds for every smooth and compactly supported function $f : G \to \mathbb{R}$. Let q_G be the infimum of $q_{G,v}$ over all choices of v as above.*

For $G = \mathbb{H}^3$, there is only one choice of v up to a sign, and we have $q_{\mathbb{H}^3} = 4$ with the exponent range satisfying (26) being $[4,\infty)$ [NY22]. For $G = \mathbb{H}^{2k+1}$ with $k \geq 2$, there is again only one choice of v up to a sign, and $q_{\mathbb{H}^{2k+1}} = 2$ with the exponent range satisfying (26) being $[2,\infty)$ [NY18].

For a general nilpotent Lie group G, we wish to know whether there are finite exponents q satisfying (26), since then not only would we reprove [EBGLD+23], but we would also have analogous quantitative nonembeddability statements: whenever q satisfies (26), then, similarly to Theorem 1, if G is Riemannian, we would have

$$c_1(B_r) \gtrsim_{G,q} (\log r)^{1/q}, \quad r \geq 2,$$

and similarly to Theorem 19, if G is Riemannian or sub-Riemannian and if N_{r_1,r_2} is an r_2-covering of B_{r_1}, where $r_1, r_2 > 0$ with $r_1 \geq 2r_2$, and with the extra condition that $r_1, r_2 \geq 1$ if $v \in \operatorname{span}\{X_1,\cdots,X_k\}$, then

$$c_1\left(N_{r_1,r_2}\right) \gtrsim_{G,q} \left(\log(r_1/r_2)\right)^{1/q}.$$

Also, if Γ is a not virtually abelian finitely generated group of polynomial growth that is quasi-isometric to G, we would have

$$c_1\left(B_n^\Gamma\right) \gtrsim_{\Gamma,q} (\log n)^{1/q}, \quad n \geq 2.$$

By the co-area formula it suffices to prove (26) when f is an indicator of a measurable set $A \subseteq G$,

The supporting Heisenberg endpoint theorem used in the proof is the three-dimensional $q = 4$ result of Naor and Young:

open questions. We will then describe our main conceptual contribution, called a *foliated corona decomposition*, which is a new structural methodology that we introduce in the *proof of* this theorem; see Remark 1.2 and mainly Section 1.2 for an overview.

For a smooth function $f\colon \mathbb{R}^3 \to \mathbb{R}$ define $\mathsf{X}f, \mathsf{Y}f\colon \mathbb{R}^3 \to \mathbb{R}$ by setting for $h=(x,y,z)\in\mathbb{R}^3$,

$$\mathsf{X}f(h) \stackrel{\mathrm{def}}{=} \frac{\partial f}{\partial x}(h) + \frac{1}{2}y\frac{\partial f}{\partial z}(h) \qquad \text{and} \qquad \mathsf{Y}f(h) \stackrel{\mathrm{def}}{=} \frac{\partial f}{\partial y}(h) - \frac{1}{2}x\frac{\partial f}{\partial z}(h). \tag{1}$$

Also, for $t\in(0,\infty)$ define $D_{\mathsf{v}}^t f\colon \mathbb{R}^3\to\mathbb{R}$ by setting for $h=(x,y,z)\in\mathbb{R}^3$,

$$D_{\mathsf{v}}^t(h) \stackrel{\mathrm{def}}{=} \frac{f(x,y,z+t)-f(h)}{\sqrt{t}}. \tag{2}$$

Theorem 1.1. *Every compactly supported smooth function $f\colon\mathbb{R}^3\to\mathbb{R}$ satisfies*[1]

$$\left(\int_0^\infty \left(\int_{\mathbb{R}^3} |D_{\mathsf{v}}^t f(h)|\,\mathrm{d}h\right)^4 \frac{\mathrm{d}t}{t}\right)^{\frac14} \lesssim \int_{\mathbb{R}^3} \big(|\mathsf{X}f(h)|+|\mathsf{Y}f(h)|\big)\,\mathrm{d}h. \tag{3}$$

Moreover, one cannot replace the $L_4(\frac{\mathrm{d}t}{t})$ norm above by an $L_q(\frac{\mathrm{d}t}{t})$ norm for any $0<q<4$.

The second assertion (sharpness) of Theorem 1.1 resolves negatively the conjecture of [LN14b] that (3) holds with the $L_4(\frac{\mathrm{d}t}{t})$ norm in the left hand side replaced by the $L_2(\frac{\mathrm{d}t}{t})$ norm. Notwithstanding the optimality of (3), it should be noted that it was previously unknown whether such a bound holds true merely for *some* finite exponent, namely that there exists $0<p<\infty$ such that in the setting of Theorem 1.1 we have

$$\left(\int_0^\infty \left(\int_{\mathbb{R}^3} |D_{\mathsf{v}}^t f(h)|\,\mathrm{d}h\right)^p \frac{\mathrm{d}t}{t}\right)^{\frac1p} \lesssim \int_{\mathbb{R}^3} \big(|\mathsf{X}f(h)|+|\mathsf{Y}f(h)|\big)\,\mathrm{d}h. \tag{4}$$

It is simple to justify (see [NY18, Remark 4]) that if (4) holds, then the analogous bound holds for any larger exponent $P>p$.

In the rest of the note, G has nilpotency step two. Put

$$Z=[\mathfrak{g},\mathfrak{g}].$$

Then Z is central. Since W Lie-generates $\mathfrak{g}$ and the group has step two,

$$\mathfrak{g}=W+[W,W], \qquad Z=[W,W]. \tag{2}$$

For $z\in Z$, $t>0$, and $f\in L^1(G)$, set

$$A_z^f(t)=\int_G |f(g\exp(tz))-f(g)|\,\mathrm{d}g$$

and, for $1\le q<\infty$,

$$\mathcal{V}_{q,z}(f)=\left(\int_0^\infty \left(\frac{A_z^f(t)}{\sqrt{t}}\right)^q \frac{\mathrm{d}t}{t}\right)^{1/q}, \qquad \mathcal{V}_{\infty,z}(f)=\sup_{t>0}\frac{A_z^f(t)}{\sqrt{t}}.$$

A useful quantitative measure of the cost of writing z as a sum of horizontal commutators is

$$\kappa_W(z)=\inf\left\{\sum_{j=1}^m\sqrt{|a_j||b_j|} : z=\sum_{j=1}^m [a_j,b_j], \quad a_j,b_j\in W\right\}. \tag{3}$$

By (2), $\kappa_W(z)<\infty$. Notice the natural homogeneity $\kappa_W(cz)=\sqrt{|c|}\,\kappa_W(z)$.

2 Complete step-two theorem

Theorem 1 (Endpoint inequality in every step-two group)**.** *There is a universal constant $C_4 < \infty$ such that, for every connected, simply connected, step-two nilpotent Lie group G, every bracket-generating horizontal space W, every $z \in [\mathfrak{g}, \mathfrak{g}]$, and every $f \in C_c^\infty(G)$,*

$$\mathcal{V}_{4,z}(f) \le C_4\, \kappa_W(z) \int_G |\nabla_W f(g)| \,\mathrm{d}g. \tag{4}$$

Moreover, for every $q \in [4, \infty]$,

$$\mathcal{V}_{q,z}(f) \le C_q\, \kappa_W(z) \int_G |\nabla_W f(g)| \,\mathrm{d}g, \tag{5}$$

where C_q is universal apart from its dependence on q. Consequently, Ryoo's Question 38 has a positive answer for every step-two nilpotent Lie group, for every allowed final-layer vector, and for every $q \ge 4$.

The proof uses the sharp $q = 4$ theorem of Naor and Young on the three-dimensional Heisenberg group.

Lemma 2 (Transfer from an elementary Heisenberg subgroup)**.** *Let $a, b \in W$ and $z = [a, b] \neq 0$. Then*

$$\mathcal{V}_{4,z}(f) \le C_{\mathrm{NY}} \int_G \big(|af(g)| + |bf(g)|\big) \,\mathrm{d}g \tag{6}$$

for every $f \in C_c^\infty(G)$, where C_{NY} is the universal constant in the three-dimensional Heisenberg theorem. In particular,

$$\mathcal{V}_{4,z}(f) \le 2C_{\mathrm{NY}} \sqrt{|a||b|} \int_G |\nabla_W f(g)| \,\mathrm{d}g. \tag{7}$$

Proof. The vectors a, b, z are linearly independent. Indeed, if the images of a and b in $\mathfrak{g}/Z$ were dependent, then $[a, b] = 0$. Hence

$$\mathfrak{h} = \operatorname{span}\{a, b, z\}$$

is a three-dimensional Heisenberg Lie algebra with $[a, b] = z$. Since the exponential map of a simply connected nilpotent group is a diffeomorphism, the connected subgroup $H = \exp(\mathfrak{h})$ is closed: it is the image under a homeomorphism of the closed linear subspace $\mathfrak{h} \subseteq \mathfrak{g}$.

Both G and H are unimodular. Weil's quotient integration formula therefore gives a G-invariant measure on G/H such that

$$\int_G \Phi(g) \,\mathrm{d}g = \int_{G/H} \int_H \Phi(xu) \,\mathrm{d}u \,\mathrm{d}\bar{x} \tag{8}$$

for compactly supported measurable Φ. For a coset representative x, define $F_x(u) = f(xu)$. The Lie algebra isomorphism from the standard Heisenberg algebra that sends its standard basis (X, Y, Z_0) to (a, b, z) identifies the Naor–Young inequality with

$$\left(\int_0^\infty \left(\frac{\int_H |f(xu\exp(tz)) - f(xu)| \,\mathrm{d}u}{\sqrt{t}} \right)^4 \frac{\mathrm{d}t}{t} \right)^{1/4} \le C_{\mathrm{NY}} \int_H \big(|af(xu)| + |bf(xu)|\big) \,\mathrm{d}u. \tag{9}$$

A scalar change in the normalization of Haar measure multiplies both sides of (9) by the same factor. Also, the left-invariant derivatives of F_x in the directions a, b are exactly the ambient derivatives af, bf at xu.

Let

$$B_x(t) = \int_H |f(xu\exp(tz)) - f(xu)|\,\mathrm{d}u.$$

By (8), $A_z^f(t) = \int_{G/H} B_x(t)\,\mathrm{d}\bar{x}$. Minkowski's integral inequality in $L^4((0,\infty), \mathrm{d}t/t)$, followed by (9) and (8), gives (6).

For $\lambda > 0$, the same central vector has the factorization

$$z = [\lambda a, \lambda^{-1} b].$$

Applying (6) to this pair and using $|cf| \le |c|\,|\nabla_W f|$ for $c \in W$, we obtain

$$\mathcal{V}_{4,z}(f) \le C_{\mathrm{NY}}\big(\lambda|a| + \lambda^{-1}|b|\big) \int_G |\nabla_W f|\,\mathrm{d}g.$$

Minimization in λ yields (7). □

Proof of theorem 1. Fix a representation $z = \sum_{j=1}^m [a_j, b_j]$. Set $z_j = [a_j, b_j]$ and $w_j = z_1 + \cdots + z_j$, with $w_0 = 0$. Since Z is central,

$$\exp(tz) = \prod_{j=1}^m \exp(tz_j).$$

For each $t > 0$, telescoping and right invariance of Haar measure give

$$\begin{aligned} A_z^f(t) &\le \sum_{j=1}^m \int_G |f(g\exp(tw_j)) - f(g\exp(tw_{j-1}))|\,\mathrm{d}g \\ &= \sum_{j=1}^m A_{z_j}^f(t). \end{aligned}$$

Minkowski's inequality and theorem 2 imply

$$\mathcal{V}_{4,z}(f) \le 2C_{\mathrm{NY}} \left(\sum_{j=1}^m \sqrt{|a_j||b_j|} \right) \int_G |\nabla_W f|\,\mathrm{d}g.$$

Taking the infimum over all factorizations proves (4).

For completeness, we record the deterministic endpoint. In a step-two group,

$$\exp(ra)\exp(ub)\exp(-ra)\exp(-ub) = \exp(ru[a,b]).$$

Thus $\exp(t[a,b])$ can be joined to the identity by a horizontal commutator loop of length at most $4\sqrt{t|a||b|}$, after optimizing over $ru = t$. Concatenating these loops and taking the infimum in (3) gives

$$d_G(e, \exp(tz)) \le 4\sqrt{t}\,\kappa_W(z). \tag{10}$$

If γ is a horizontal path from e to $\exp(tz)$, differentiation of $f(g\gamma(\cdot))$, followed by Fubini and right invariance of Haar measure, yields

$$A_z^f(t) \le \mathrm{length}(\gamma) \int_G |\nabla_W f|\,\mathrm{d}g.$$

Together with (10), this proves

$$\mathcal{V}_{\infty,z}(f) \le 4\kappa_W(z) \int_G |\nabla_W f| \,\mathrm{d}g. \tag{11}$$

Interpolation of the scalar function $t \mapsto A_z^f(t)/\sqrt{t}$ between $L^4(\mathrm{d}t/t)$ and $L^\infty(\mathrm{d}t/t)$ proves (5). $\square$

Corollary 3 (The supplied Carnot statement)**.** *If G is a step-two Carnot group with stratification $\mathfrak{g} = V_1 \oplus V_2$, then theorem 1 applies with $W = V_1$ and every $z \in V_2$. In particular, the main theorem in the supplied packet is correct.*

Corollary 4 (Discrete consequence)**.** *Combining theorem 1 with Ryoo's discretization argument gives exponent $q = 4$ in the corresponding discrete endpoint inequality for finitely generated step-two nilpotent groups, and hence for the step-two cases covered after passing to finite-index and finite-torsion modifications in Ryoo's framework.*

3 A sufficient mechanism for exponent two

The exponent four is not always optimal. Naor and Young proved exponent two for $\mathbb{H}^{2k+1}$ when $k \ge 2$. The same coset-transfer argument gives the following useful sufficient criterion.

Proposition 5 (Higher-dimensional Heisenberg transfer)**.** *Suppose that a central vector $z \in [\mathfrak{g}, \mathfrak{g}]$ generates the center of a closed connected subgroup $H \le G$ whose Lie algebra is isomorphic to $\mathfrak{h}^{2k+1}$ for some $k \ge 2$, and suppose that the horizontal layer of H is contained in W. Then*

$$\mathcal{V}_{2,z}(f) \lesssim_{H,W} \int_G |\nabla_W f| \,\mathrm{d}g.$$

Consequently the same estimate holds for every $q \ge 2$. More generally, if a vector v is a finite sum of central vectors satisfying this criterion, then the conclusion holds for v by central telescoping.

Proof. Disintegrate Haar measure over G/H, apply the higher-dimensional Heisenberg $q = 2$ theorem on each coset, and use Minkowski exactly as in theorem 2. Central telescoping handles finite sums, and interpolation with (11) gives all larger exponents. $\square$

4 Verification of the supplied proof

The supplied argument has the right structure and no fatal gap. The following are the points that require, or benefit from, explicit wording.

1. The subgroup generated by $a, b, [a, b]$ is closed. In a simply connected nilpotent group this follows immediately from the fact that the exponential map is a diffeomorphism.
2. Weil disintegration is applicable because the subgroup is closed and both groups are unimodular.
3. For $F_x(u) = f(xu)$, the subgroup left-invariant derivatives are the ambient left-invariant derivatives at xu.
4. Haar normalization under the Lie-group identification with $\mathbb{H}^3$ is harmless because both sides of the endpoint inequality are homogeneous of degree one in Haar measure.

5. The final telescoping identity uses centrality and right invariance of Haar measure. Nilpotent groups are unimodular, so this is valid.

The principal strengthening over the packet is that no grading or Carnot dilation is used in the proof. The only algebraic inputs are step two and the fact that the chosen horizontal space bracket-generates the Lie algebra.

5 Literature status as of June 18, 2026

Ryoo's revised paper states the endpoint problem as Questions 38 and 42 and records only the exact Heisenberg answers: exponent range $[4, \infty)$ in $\mathbb{H}^3$, and $[2, \infty)$ in $\mathbb{H}^{2k+1}$, $k \geq 2$. It then formulates the existence of a finite exponent in every nilpotent group as a conjecture. The article was accepted in March 2026 and made available online in April 2026; the endpoint questions remain in the revised version.

We searched exact phrases from Questions 38 and 42 and Conjecture 43, the arXiv record and papers citing it, combinations of "endpoint", "vertical perimeter", "nilpotent", "Carnot", and "L^1", and recent 2025–2026 Carnot-group BV and nonlocal Sobolev papers. The search located the Heisenberg theorems, the qualitative theorem that general nonabelian nilpotent groups do not embed bi-Lipschitzly into L^1, and recent isotropic or horizontal nonlocal characterizations of BV functions. None supplies the one-directional, last-layer, finite-q estimate in (1), and no counterexample to it was located. Thus, on the evidence found, there is no known full answer beyond the Heisenberg cases and the complete step-two upper bound proved above.

This is necessarily a literature-search conclusion rather than a theorem: absence from the sources searched cannot certify absolute nonexistence of an unpublished or differently phrased result.

6 Why the elementary proof does not iterate to higher step

It is useful to isolate the precise obstruction. Suppose $v = [a, w]$ lies in the top layer of a step-s group, where $a \in W$ and w has homogeneous degree $s - 1$. The subgroup generated by a, w, v is three-dimensional Heisenberg, because v is central. Applying the Heisenberg theorem there, however, introduces the derivative wf, which is not controlled by the first-order horizontal variation of f.

At the level of finite differences one has the exact commutator path estimate

$$A_v^f(ru) \leq 2A_a^f(r) + 2A_w^f(u). \tag{12}$$

Writing $t = ru$, choosing $r = \lambda t^{1/s}$, and formally normalizing the w-term at its degree gives

$$\frac{A_v^f(t)}{t^{1/s}} \lesssim \lambda \int_G |\nabla_W f| + \lambda^{-1/(s-1)} \frac{A_w^f(\lambda^{-1} t^{(s-1)/s})}{(\lambda^{-1} t^{(s-1)/s})^{1/(s-1)}}.$$

This suggests a distribution-function induction and even predicts finite candidate exponents. The induction does not close: w is generally not central in G, while it becomes central only after quotienting by the top layer. A section of that quotient introduces a top-layer cocycle into the horizontal derivatives, precisely the quantity one is trying to estimate. In the Engel group this obstruction is already present with $v = [X, [X, Y]]$ and $w = [X, Y]$.

Other approaches examined include nested commutator words, quotient induction, constant-normal sets in the Engel and free rank-two step-three groups, and recent nonlocal BV characterizations. They give useful pointwise or qualitative information, but no scale-summable estimate and no counterexample. In particular, the known pathological constant-normal examples have top-central translation moduli that decay faster than the critical $t^{1/s}$ normalization in the model calculations, so they do not by themselves disprove the conjecture.

7 Conclusion

The partial result is correct and can be upgraded to a complete theorem for all step-two nilpotent Lie groups with arbitrary bracket-generating left-invariant Carnot–Carathéodory structures. The proof is short because every commutator vector is a finite sum of centers of genuine horizontal three-dimensional Heisenberg subgroups. No analogous reduction is available in higher step: the second generator of the elementary Heisenberg subgroup is then a nonhorizontal lower-central direction. We did not obtain a rigorous higher-step proof or counterexample, and none was found in the literature search.

References

[1] S.-Y. Ryoo, *Quantitative nonembeddability of groups of polynomial growth into uniformly convex spaces*, J. Funct. Anal. (2026), article 111491; arXiv:2207.11305v2.

[2] A. Naor and R. Young, *Foliated corona decompositions*, Acta Math. **229** (2022), 55–200.

[3] A. Naor and R. Young, *Vertical perimeter versus horizontal perimeter*, Ann. of Math. (2) **188** (2018), 171–279.

[4] S. Eriksson-Bique, C. Gartland, E. Le Donne, L. Naples, and S. Nicolussi Golo, *Nilpotent groups and bi-Lipschitz embeddings into L^1*, Int. Math. Res. Not. IMRN (2023), no. 12, 10759–10797.

[5] F. Serra Cassano and K. Zambanini, *Characterizations of Sobolev and BV functions on Carnot groups*, arXiv:2603.28996, 2026.

[6] C. Bellettini and E. Le Donne, *Sets with constant normal in Carnot groups: properties and examples*, Comment. Math. Helv. **96** (2021), 149–198.